ALAGARSAMY DURAIKUMARAN | Singaporean | GitHub | LinkedIn

EDUCATION

Nanyang Technological University (NTU), Singapore Aug 2024 - Dec 2027
Bachelor of Engineering (Honours) in Computer Engineering
Relevant Modules - Algorithm Design and Analysis, Software Engineering, Data Structures and Algorithm

WORK EXPERIENCE

TATA Consultancy Services (TCS), AI Powered Research & Innovation Centre (Pace Port), Singapore May 2026 - Present
Generative AI Engineer Intern

- Architected a 7-agent AI pipeline on Microsoft Azure using Azure AI Foundry for a multinational marine enterprise's finance operations, automating complex workflows and delivering actionable insights via Power BI while strictly enforcing data governance.
- Built an autonomous Voice AI assistant orchestrating 16 MCP servers with Google Gemini, demonstrating real-time conversational intelligence and multi-tool execution for 200+ attendees at the SG Youth Forum.
- Engineered Campfire, a voice-activated Claude multi-agent application builder and leveraged it to lead live "vibe coding" workshops, translating complex generative workflows for 200+ TATA executives, students, and industry leaders.
- Integrated diverse, large-scale datasets to author a quantum research report on hybrid quantum-classical transformers, driving strategic technical engagements with 4 key ecosystem partners in Singapore's quantum sector.
- Designed and demoed CareerQuest, an interactive educational pathway application leveraging the SkillsFuture dataset, to 200+ students at the SG Youth Forum.
- Built an AI-augmented business management simulation game for an NGO client, integrating real time analytics to train organizational leaders on data-driven decision making, strategic resource allocation and their company's core values.
- Engineered POC Hub, a centralised client facing enterprise platform showcasing TCS Pace Port's AI portfolio, accelerating technical discovery and streamlining solution proposals during executive corporate visits.

SKILLS

- AI/ML: LLMs, LangChain, LangGraph, RAG pipelines, Prompt Engineering, Claude Code, CoPilot, Predictive Modelling
- Cloud & Backend: Python, SQL, AWS, GCP, Microsoft Azure (AI Foundry), DataBricks
- Frontend: React, Vite, Tailwind CSS, JavaScript, HTML, CSS

PROJECTS

Quantitative Stock Signal Engine & Backtesting Platform | GitHub Jul 2026 - Jul 2026

- Developed an algorithmic scanner with a custom built mathematical indicator suite (RSI, MACD) and NLP sentiment analysis, reducing false positive trade signals through multi-layer market regime filtering.
- Architected a rigorous historical backtesting engine processing 5 years of financial data, validated by 54 automated tests, to compute forward-return distributions and prove statistical edge prior to live deployment.

AssetVault: AI-Powered Financial Data & Property Management Platform | GitHub Aug 2026 - Aug 2026

- Developed a privacy compliant financial asset management platform enforcing strict account based data isolation, directly demonstrating enterprise-grade data governance and risk management capabilities.
- Integrated OCR and LLM pipelines to autonomously parse unstructured financial documents, powering a centralized analytics dashboard that tracks predictive ROI trends and accelerates manual data entry workflow by 80%.

SAP x **NTU PEAK Program (Industry Project)** Aug 2025 - Oct 2025

- Selected for NTU PEAK industry program with SAP and proposed SAP Matrix, a solution concept integrated with Joule AI Agent.
- Delivered recommendations reducing carbon footprint estimation errors by 15% and material waste by 20% through AI-driven predictive logistics and supplier auditing.

HACKATHONS

Deep Learning Week 2026 Hackathon (Finalist) - CrashGuard AI | GitHub | Demo Video Feb 2026 - Mar 2026

- Built an emergency intelligence web application using React, JavaScript, Python, TensorFlow Lite, and Ollama, enabling real-time crash detection, AI incident analysis, and automated emergency reporting through an end-to-end vision and LLM pipeline.
- Improved emergency response readiness by generating structured crash assessment reports and automatically notifying emergency contacts/dispatchers via EmailJS, reducing manual reporting time during accidents.

Microsoft x NTU Product Club Hackathon (2nd Place) – PromptHub Jun 2025 - Jun 2025

- Built PromptHub, an AI prompt marketplace developed during the Microsoft x NTU Product Club hackathon, enabling users to create, discover, and sell high-quality prompts for LLM workflows.
- Improved prompt accessibility and monetisation by implementing searchable prompt libraries and creator listings, reducing time spent crafting effective prompts for AI tools.

LEADERSHIP EXPERIENCE

NTU Students' Union, NTUSU Sep 2024 - Aug 2025
CCA Liaison Director

- Coordinated 200+ CCAs and streamlined operations while leading the annual Welcome Week for 5,000+ students, enhancing campus engagement.